

Core TensorFlow		
Sub-module	What it does	Key API
tensorflow (root)	High-level utilities, constants, data types.	tf.constant, tf.Variable, tf.Tensor, tf.shape, tf.cast, tf.reshape
tensorflow.math	Mathematical ops (element-wise, linear algebra).	tf.add, tf.matmul, tf.linalg.solve, tf.exp, tf.log
tensorflow.random	Random number generation.	tf.random.normal, tf.random.uniform, tf.random.shuffle
tensorflow.linalg	Linear algebra (matrix ops).	tf.linalg.inv, tf.linalg.eig, tf.linalg.svd
tensorflow.image	Image processing.	tf.image.resize, tf.image.flip_left_right, tf.image.random_flip_up_down
tensorflow.signal	Signal processing (FFT, filtering).	tf.signal.fft, tf.signal.filter_biquad
tensorflow.io	File I/O, data pipelines.	tf.io.read_file, tf.io.decode_csv, tf.data.TFRecordDataset
tensorflow.summary	Logging for TensorBoard.	tf.summary.scalar, tf.summary.image, tf.summary.flush
tensorflow.config	Runtime configuration (GPU, logging).	tf.config.list_physical_devices, tf.debugging.set_log_device_placement
TensorFlow Keras		
tensorflow.keras (root)	High-level API for building models.	tf.keras.Sequential, tf.keras.Model
tensorflow.keras.layers	Pre-built layers.	Dense, Conv2D, LSTM, BatchNormalization, Dropout
tensorflow.keras.models	Model utilities.	tf.keras.models.load_model, tf.keras.models.save_model
tensorflow.keras.losses	Loss functions.	CategoricalCrossentropy, MeanSquaredError, SparseCategoricalCrossentropy
tensorflow.keras.metrics	Evaluation metrics.	Accuracy, Precision, Recall, AUC
tensorflow.keras.optimizers	Optimizers.	Adam, SGD, RMSprop, Adagrad
tensorflow.keras.callbacks	Training callbacks.	ModelCheckpoint, EarlyStopping, TensorBoard, LearningRateScheduler
tensorflow.keras.utils	Helpers.	to_categorical, plot_model, get_file
tensorflow.keras.applications	Pre-trained models.	ResNet50, VGG16, InceptionV3
TensorFlow Datasets		
tensorflow.data (root)	Data pipeline building.	tf.data.Dataset.from_tensor_slices, Dataset.map, Dataset.batch, Dataset.prefetch
tensorflow.data.experimental	Experimental ops (e.g., map_and_batch, interleave).	tf.data.experimental.AUTOTUNE
tensorflow.keras.preprocessing	Legacy data generators.	ImageDataGenerator, TextVectorization
tensorflow.data.experimental.CsvDataset	CSV streaming.	tf.data.experimental.CsvDataset
tensorflow.data.Dataset	Dataset class itself.	tf.data.Dataset.zip, Dataset.concatenate, Dataset.interleave
TensorFlow Estimators		
tensorflow.estimator (root)	High-level Estimator API.	tf.estimator.Estimator, tf.estimator.inputs.numpy_input_fn
tensorflow.estimator.BoostedTreesClassifier	Gradient-boosted trees.	tf.estimator.BoostedTreesClassifier
tensorflow.estimator.DNNClassifier	DNN classifiers.	tf.estimator.DNNClassifier
tensorflow.estimator.LinearClassifier	Linear models.	tf.estimator.LinearClassifier
tensorflow.estimator.export	Exporting models for serving.	estimator.export_savedmodel, tf.saved_model.save
TensorFlow Lite		
tensorflow.lite (root)	Core TFLite APIs.	tf.lite.TFLiteConverter.from_saved_model
tensorflow.lite.python.interpreter	Interpreter for inference.	tf.lite.Interpreter, interpreter.allocate_tensors()
tensorflow.lite.python.gen_tensor	Helper for tensor creation.	tf.lite.Tensor
tensorflow.lite.python.anon	Utilities for quantization.	tf.lite.TFLiteConverter.optimizations
TensorFlow Federated		
tensorflow_federated (root)	Federated learning framework.	tff.learning.build_collaborative_keras_model
tensorflow_federated.simulation	Simulation utilities.	tff.simulation.datasets.emnist.load_data
tensorflow_federated.learning	Training APIs.	tff.learning.from_keras_model, tff.learning.build_federated_averaging_process
tensorflow_federated.program	Programmatic workflows.	tff.program.run
TensorFlow Probability		
tensorflow_probability (root)	Probabilistic models & distributions.	tfd.Normal, tfd.Categorical
tensorflow_probability.distributions	Discrete & continuous distributions.	tfd.Bernoulli, tfd.MultivariateNormalTriL
tensorflow_probability.bijectors	Transformations.	tfd.TransformedDistribution, tfb.SoftmaxCentered
tensorflow_probability.mcmc	MCMC samplers.	tfp.mcmc.HamiltonianMonteCarlo, tfp.mcmc.sample_chain
TensorFlow Addons		
tensorflow_addons (root)	Community-maintained extensions.	tfa.losses.SoftmaxCrossEntropyWithLogits
tensorflow_addons.layers	Extra layers.	tfa.layers.SpectralNormalization, tfa.layers.Conv2DTranspose
tensorflow_addons.optimizers	Advanced optimizers.	tfa.optimizers.AdamW, tfa.optimizers.Lookahead
tensorflow_addons.metrics	Metrics.	tfa.metrics.F1Score, tfa.metrics.AUC
tensorflow_addons.image	Image ops.	tfa.image.random_rotate, tfa.image.rotate
tensorflow_addons.text	NLP utilities.	tfa.text.Tokenizer, tfa.text.WordPieceTokenizer
TensorFlow Hub		
tensorflow_hub (root)	Loading reusable modules.	hub.load, hub.KerasLayer
tensorflow_hub.Module	Represents a TF Hub module.	Module.signatures, Module.save
tensorflow_hub.KerasLayer	Wraps a Hub module as a Keras layer.	tf.keras.layers.Dense, hub.KerasLayer